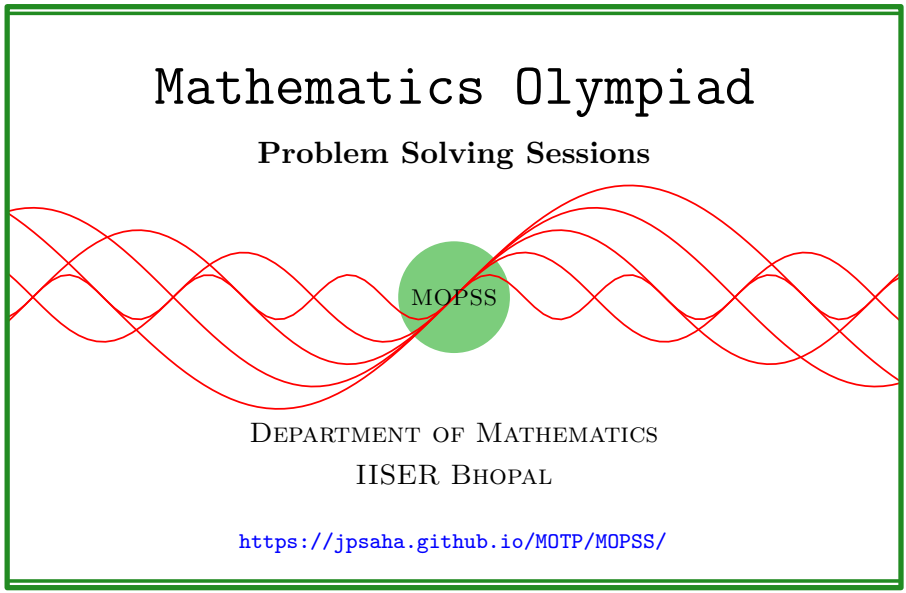
Circles



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

1 Exercise (RMO 1994 P6,) 2
2 Exercise (RMO 2006 P5,) 2
3 Exercise (RMO 2013d P1,) 2
4 Exercise (RMO 2013d P5,) 2
5 Exercise (RMO 2013e P1,) 2
6 Exercise (RMO 2014d P1,) 2
7 Exercise (INMO 2015 P1,) 2

§1 Circles

Exercise 1 (RMO 1994 P6,). Let AC and BD be two chords of a circle with center O such that they intersect at right angles inside the circle at the point M. Suppose K and L are the mid-points of the chord AB and CD respectively. Prove that $OKML$ is a parallelogram.

Exercise 2 (RMO 2006 P5,). Let $ABCD$ be a quadrilateral in which AB is parallel to CD and perpendicular to AD; $AB = 3CD$; and the area of the quadrilateral is 4. If a circle can be drawn touching all the sides of the quadrilateral, find its radius.

Exercise 3 (RMO 2013d P1,). Let ω be a circle with centre O. Let γ be another circle passing through O and intersecting ω at points A and B. A diameter CD of ω intersects γ at a point P different from O. Prove that $\angle APC = \angle BPD$.

Exercise 4 (RMO 2013d P5,). In a triangle ABC, let H denote its orthocentre. Let P be the reflection of A with respect to BC. The circumcircle of triangle ABP intersects the line BH again at Q, and the circumcircle of triangle ACP intersects the line CH again at R. Prove that H is the incentre of triangle PQR.

Exercise 5 (RMO 2013e P1,). Let ABC be an isosceles triangle with $AB = AC$ and let Γ denote its circumcircle. A point D is on the arc AB of Γ not containing C and a point E is on the arc AC of Γ not containing B such that $AD = CE$. Prove that BE is parallel to AD.

Exercise 6 (RMO 2014d P1,). Let $ABCD$ be an isosceles trapezium having an incircle; let AB and CD be the parallel sides and let CE be the perpendicular from C on to AB. Prove that CE is equal to the geometric mean of AB and CD.

Exercise 7 (INMO 2015 P1,). Let ABC be a right-angled triangle with $\angle B = 90°$. Let BD is the altitude from B on AC. Let P, Q and I be the incenters of triangles ABD, CBD and ABC respectively. Show that circumcenter of triangle PIQ lie on the hypotenuse AC.

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)